

Founding Engineer and current Staff Software Engineer at hiop. Built the core distributed data platform from zero and scaled engineering from 5 to 15+. 5+ years on high-throughput systems and cloud-native data infrastructure, with measurable impact: 50% query latency reduction, release cycles from days to minutes. Active OSS contributor.

Experience

hiop - Founding Engineer

10/21 - Present

Staff Software Engineer (01/23 - Present):

- Built the company's core distributed data processing platform from zero: architected the system design, implemented real-time ingestion pipelines over gRPC, and delivered the scalable backbone processing thousands of data pipelines daily
- Drove a 50% reduction in query engine latency by profiling execution hot paths and rewriting performance-critical components in Rust, unlocking latency-sensitive use cases that were previously out of reach
- Operated distributed systems in production through on-call rotations, incident reviews, and capacity planning: consistently maintaining SLA targets under real-world load growth
- Architected end-to-end CI/CD automation and SDLC tooling across 100+ services, compressing release cycles from days to minutes and enabling the team to ship continuously with high confidence
- Automated the full user onboarding and resource provisioning lifecycle, eliminating ~80% of manual ops overhead and enabling customer growth without a proportional increase in headcount
- Mentored 15+ engineers and helped define the hiring bar and agile processes as the org scaled 5 to 15+

Software Engineer (10/21 - 12/22):

- Co-designed and shipped the production ML/DL inference framework, enabling product teams to deploy 10+ models to production in a single configuration step rather than weeks of custom integration work
- Developed and maintained 20+ core libraries forming the internal engineering toolchain, establishing shared abstractions and coding standards that accelerated development velocity across 15+ engineers

CNR's Institute of Biophysics

12/20 - 08/21

Junior Full Stack Developer

Developed from the ground a Python and JS-based web application named *NeuroFeatureExtract* to analyse electrophysiological features as part of the *Human Brain Project*

Others

Bioinformatics Trainee @ <i>ISMETT</i>	05/21 - 08/21
ML Research Project Collaborator @ <i>UNINA</i>	11/19 - 04/20
Full Stack Developer Trainee @ <i>Informamuse</i>	01/19 - 04/19

Open Source contributions

datafusion, arrow-rs, datafusion-python, datafusion-extra-functions, maturin-action, conda, rust-toolchain, setup-python

Skills

Stack

Languages: Rust, Python, TypeScript, Java, Scala, C
Data: Postgres, Pulsar, Arrow, DataFusion, Iceberg
Infra: AWS, GCP, Docker, Terraform, GH Actions, gRPC

Certifications

AWS: *Practitioner, Developer*
GCP: *Digital Leader*
GitHub: *Foundations, Actions, Advanced Security*

Languages

Italian: native
English: C1
French: A2

Education

Double Master in Computer Science @ UNIPA & UGE

09/19 - 10/21

Graduated cum laude with thesis on *RDF data and columnar formats*

Bachelor in Computer Science @ UNIPA

09/16 - 07/19

Graduated cum laude with thesis on *suffix array data structure*

Others

Organizer @ <i>Rust Language Milano</i>	Present
<i>CMM Pre-doctoral Research School</i>	08/20
<i>ESS Information Science School</i>	09/19
Erasmus Exchange @ <i>ELTE</i>	09/18 - 01/19
<i>Bocconi Logic Games</i> Regional Winner	04/15

Publication

The EBRAINS NeuroFeatureExtract: An Online Resource for the Extraction of Neural Activity Features From Electrophysiological Data